

SLEEP



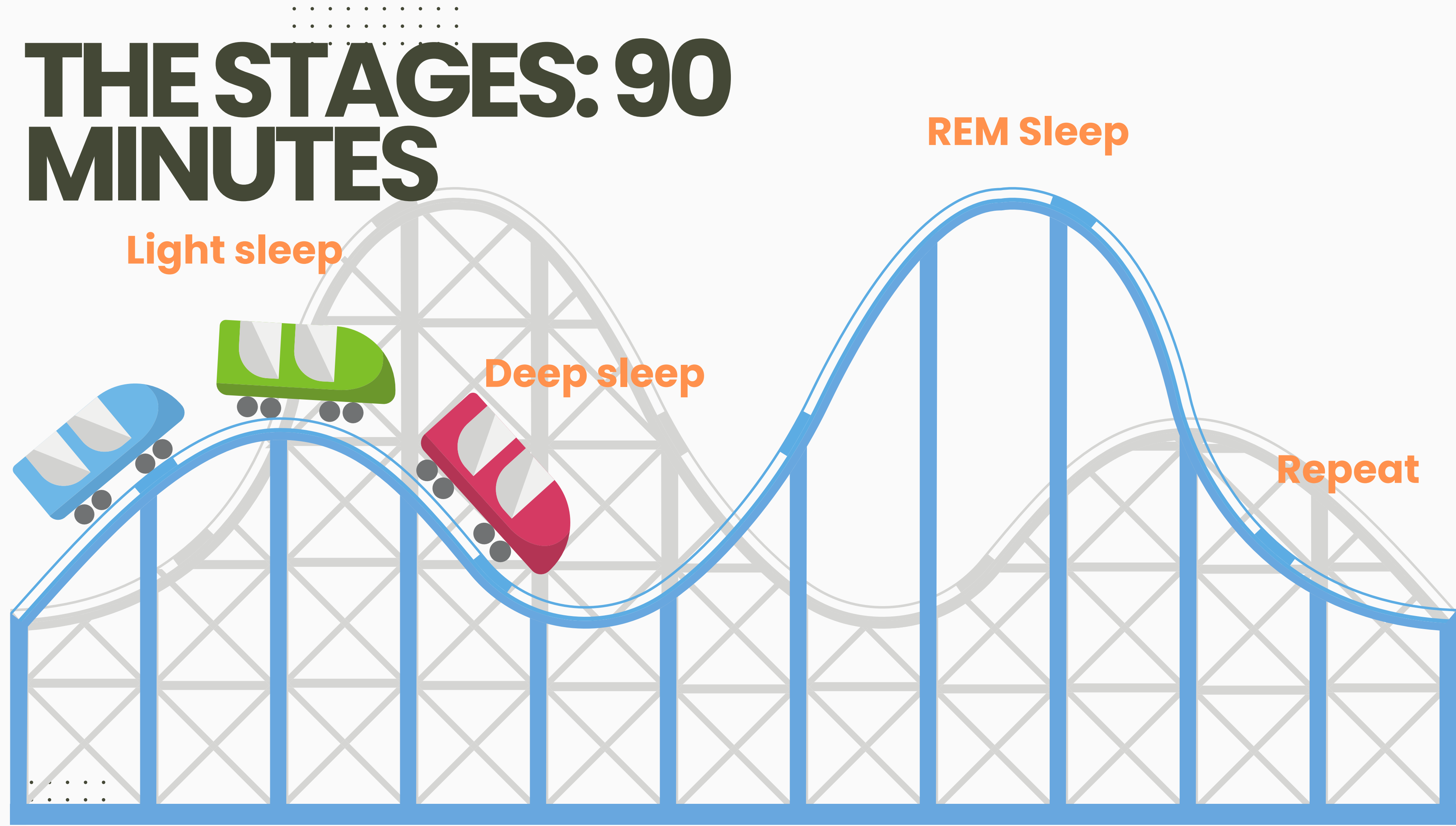
# THE STAGES: 90 MINUTES

Light sleep

Deep sleep

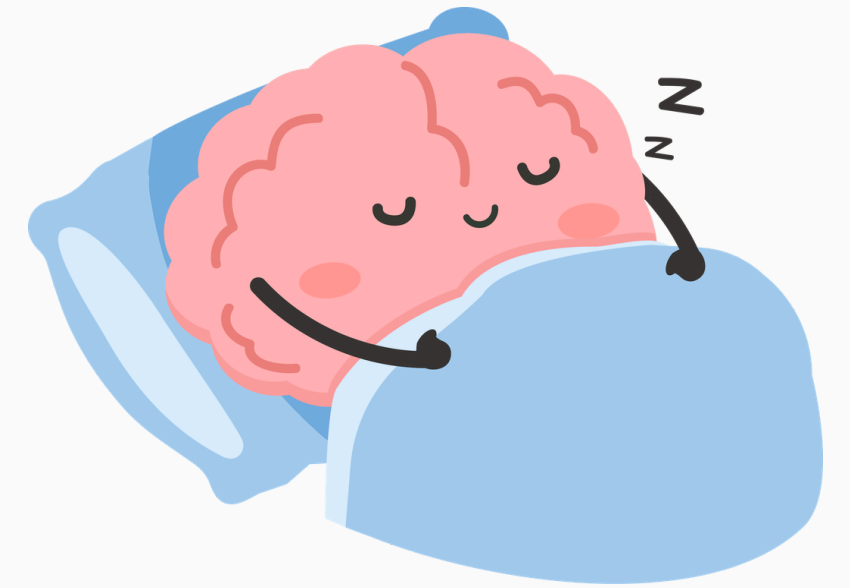
REM Sleep

Repeat



# LIGHT SLEEP

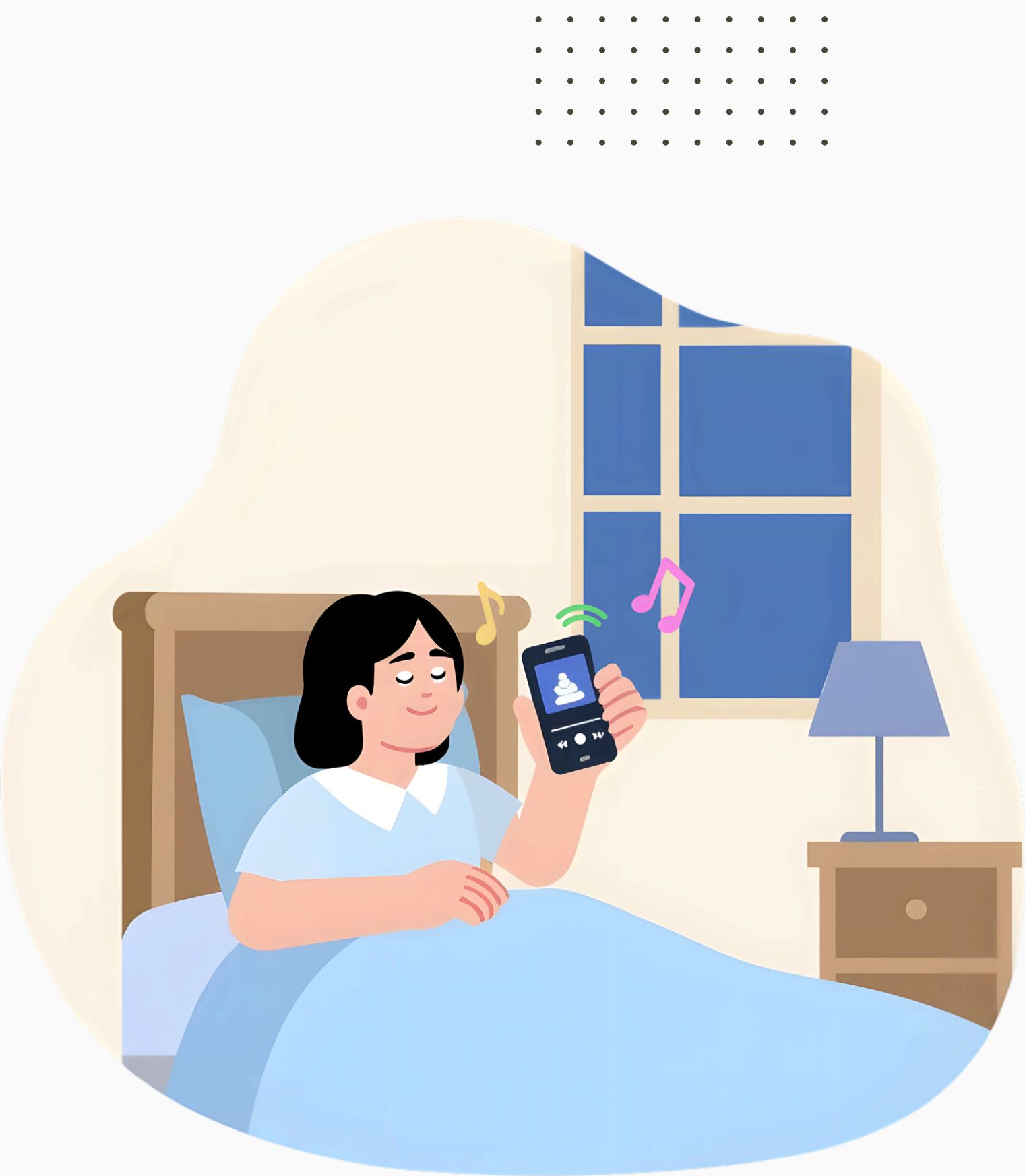
## PART 1 AND 2



- Drifting into sleep but can still be easily woken up by a quiet noise.
- Your muscles might suddenly twitch. (Falling)

### Part 2

- Heart rate slows down
- Body temperature drops
- The brain starts producing quick bursts of electrical activity called sleep spindles
  - help transfer info you learned from short term to long term
  - noise canceling from outside world



# DEEP SLEEP

- **Body is completely relaxed**
- **Goes into a mode called Slow-Wave Sleep.**
- **The body releases growth hormones to make kids grow taller**
- **Heals muscles from recess and sports**
- **Boosts the immune system to fight off colds.**
- **super hard to be woken up**





# REM SLEEP

- **Rapid Eye Movement**
- **Body is totally paralyzed**
- **Brain waves look exactly like you are wide awake.**
- **This is the stage where the Hippocampus takes everything learned during the day and cements it into long-term memory.**
- **Also when we dream**

# ACTIVITY!

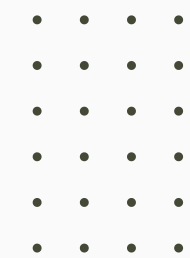
**The Goal: Prove that your brain can hold onto information even when things get chaotic—just like it does during REM sleep!**

**Step 1: You are going to see a list of 7 word pairs. You have exactly 1 minute to stare at them and try to lock them into your brain**

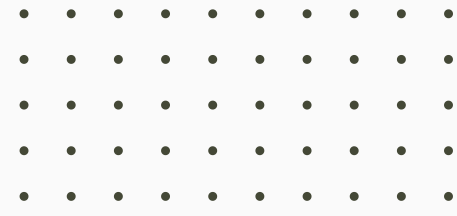
**Step 2: Do a distracting task for two minutes. No cheating or whispering the words!**

**Step 3: I will call out the first word of the pair. Your job is to write down the matching wacky word from memory.**

**Let's see who has the ultimate sleep superpower!**



# QUIZ TIME

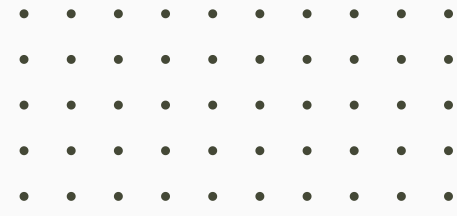


**What is the name of the tiny brain helper that acts like a mailroom clerk sorting your memories?**

- A. Hippocampus**
- B. Spine**
- C. Cerebellum**
- D. Occipital Lobe**



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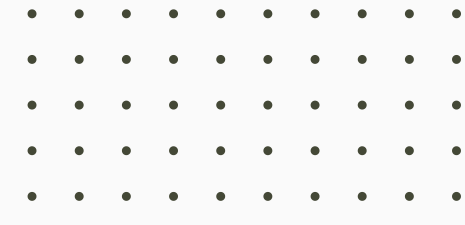
**C. Cerebellum**

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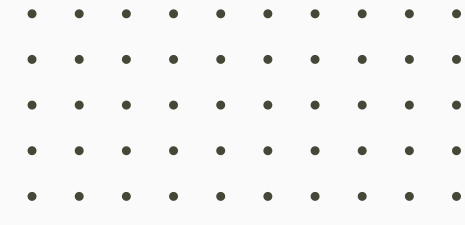


**What happens during Stage 1 of sleep or Part 1 of sleep**

- A. Your brain creates dreams**
- B. You are in deep unwakeable sleep**
- C. You're drifting into sleep and twitching**
- D. Your muscles are paralyzed**



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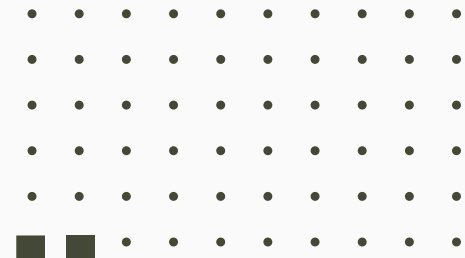
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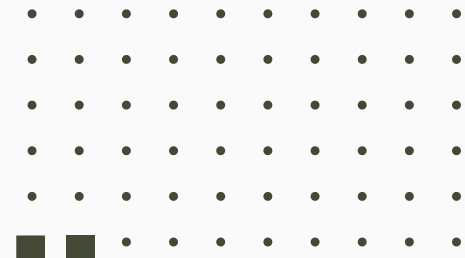


**What are sleep spindles, which occur during Stage 2 or Part 2 sleep?**

- A. Quick bursts of electrical activity which help lock in memories and noise cancel**
- B. Hormones that make you grow taller**
- C. Tiny muscles in your eyes that make you blink**
- D. Help make dreams**



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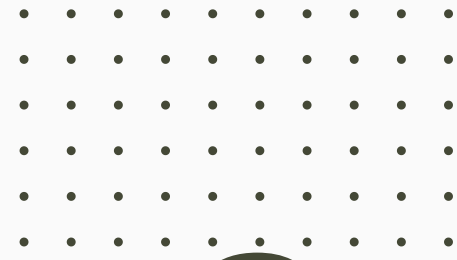
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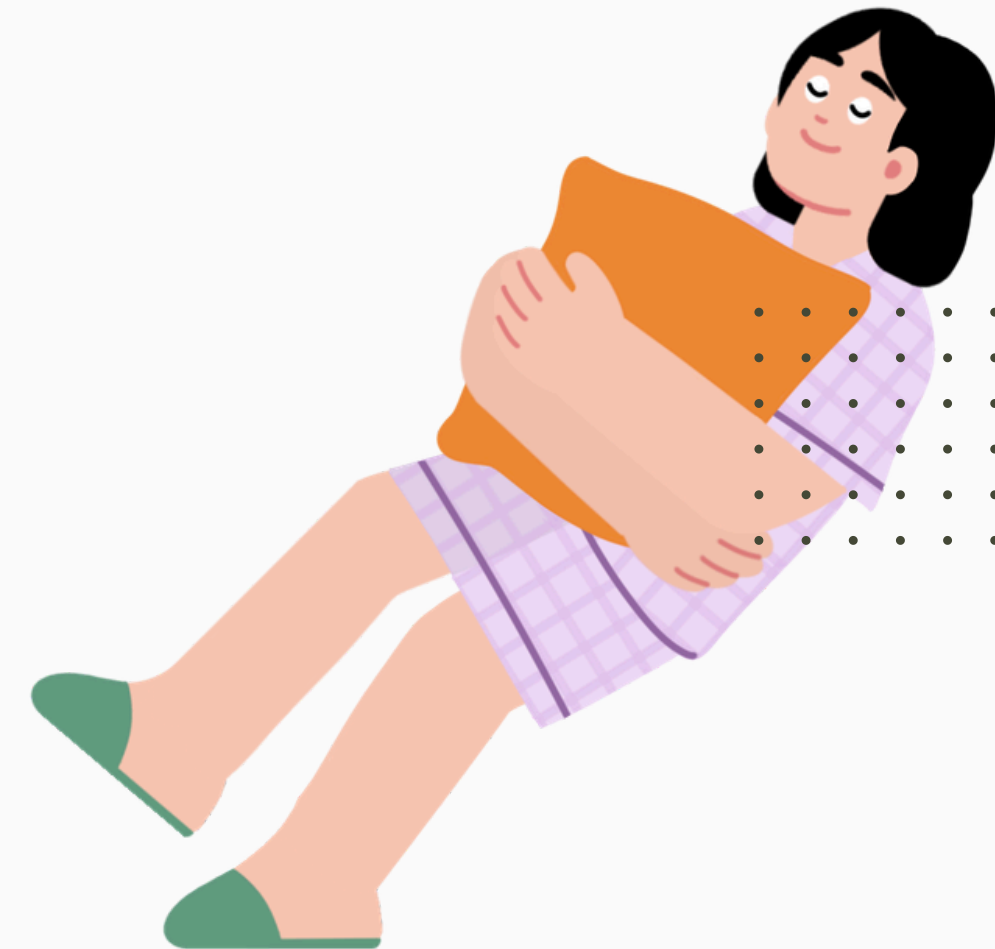
**What is Stage 3 sleep**

**A. Light sleep part 1**

**B. Light sleep part 2**

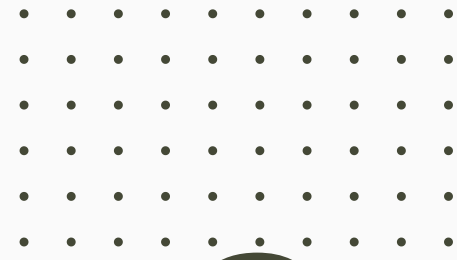
**C. Deep Sleep**

**D. REM Sleep**





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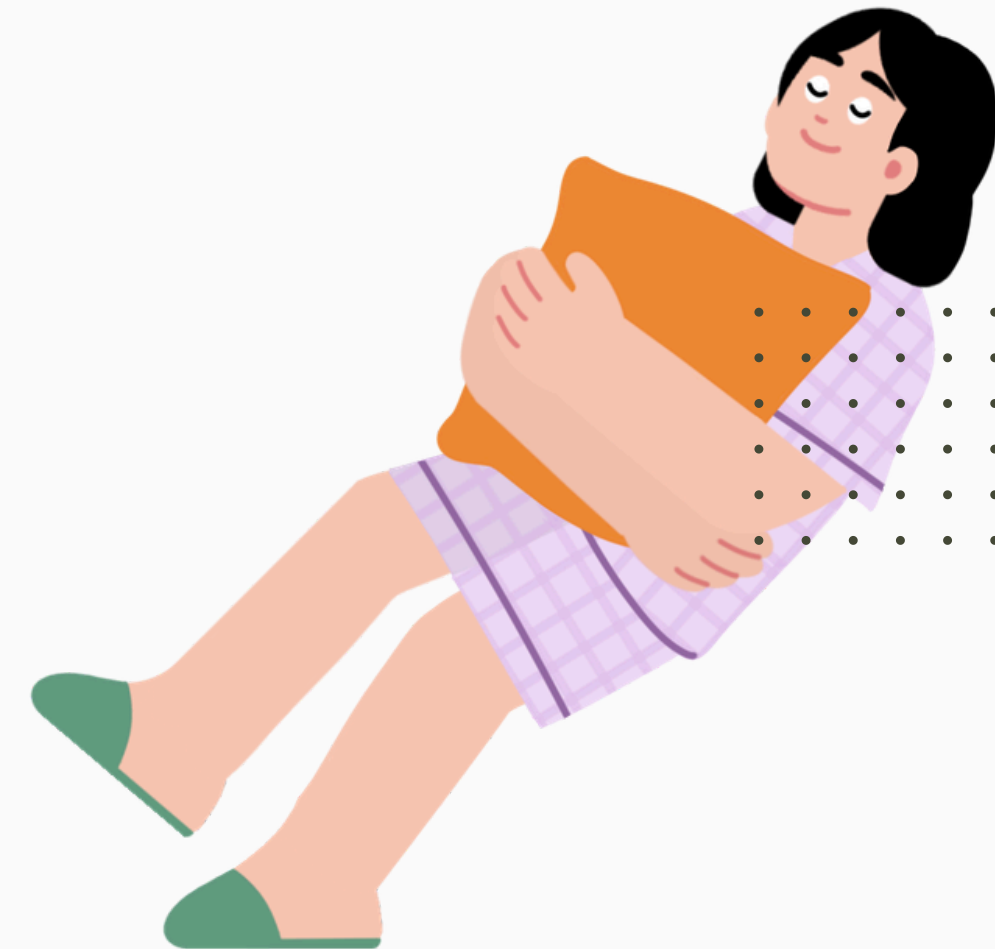
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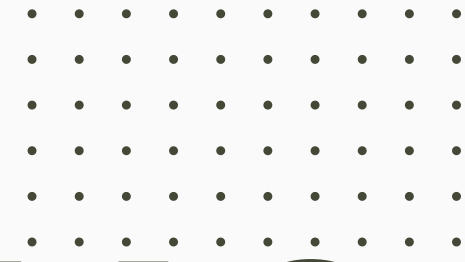


**What does REM Stand for?**

- A. Rapid Eye Movement**
- B. Relaxed Energy Mode**
- C. Resting Electrical Muscle**
- D. Reading Every Memory**



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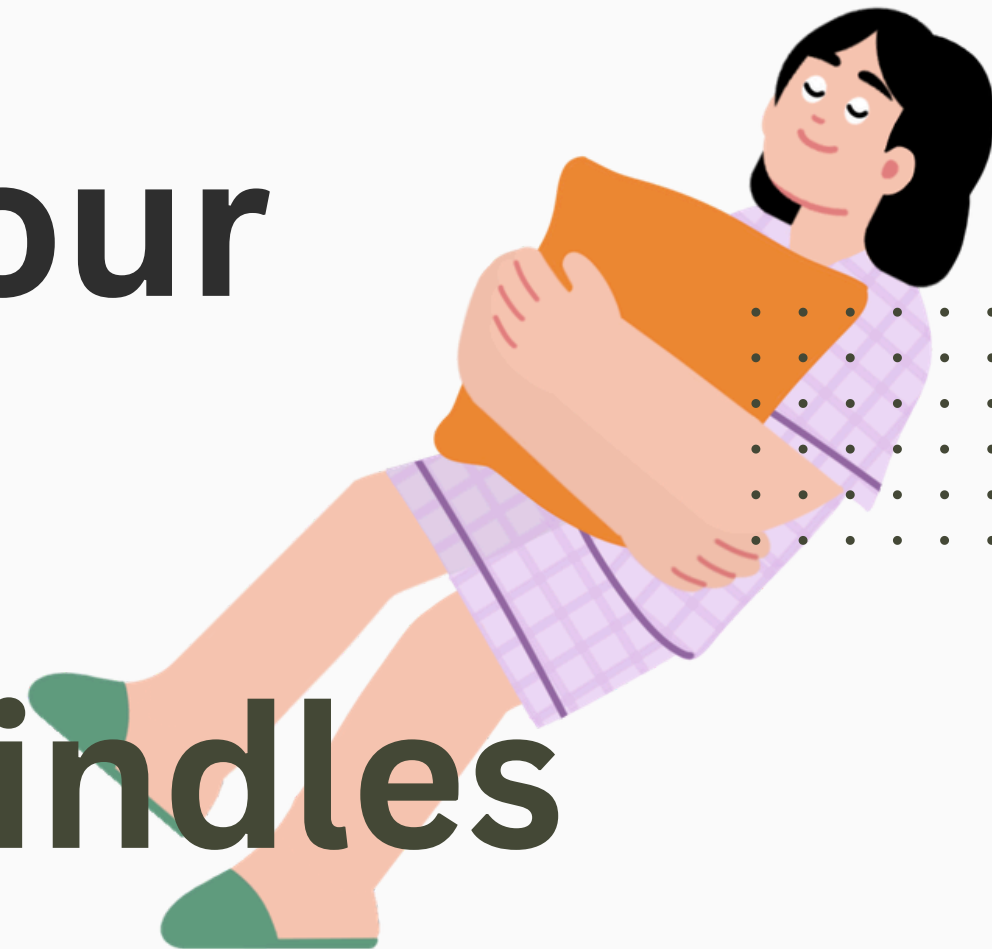
Why does your body paralyze (freeze) your muscles during REM sleep?

A. To pump blood to legs

B. To help bones grow

C. To keep you from acting your dreams

D. Your brain has no more spindles



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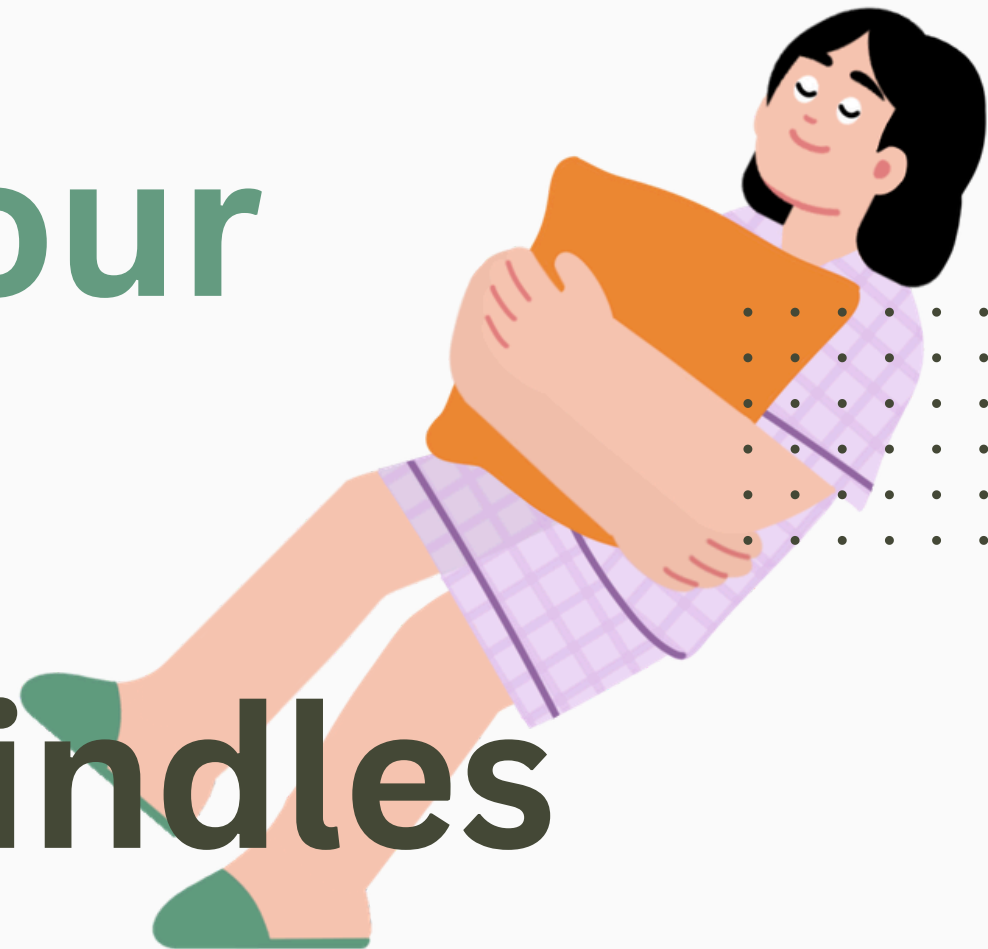
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# QUIZ TIME

If you are studying for a big test, why is a full night of sleep better than staying up late to cram?

- A. Sleep clears out your brain so you forget wrong answers
- B. Moves memories from short term to long term
- C. Staying up late means your brain grows
- D. Your brain automatically reads textbooks while you sleep





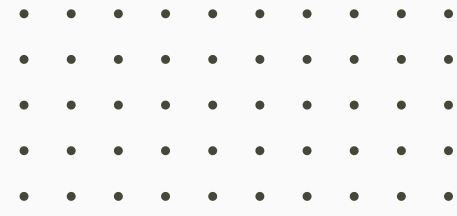
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# QUIZ TIME



**How long is one full trip through all the sleep stages (one sleep rollercoaster cycle)?**

**A. 90 minutes**

**B. 8 hours**

**C. 24 hours**

**D. 30 mins**



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**THANK  
YOU  
VERY  
MUCH**